



Fatemeh Vahid Younesi

AI Engineer

AI Engineer specializing in production NLP, LLM and speech/audio AI systems, with experience taking AI models from experimentation to real-world deployable services.

Contact

- ✉ fatemeh.vahidunesi@gmail.com
- ☎ (+98)9129346714
- 📍 Tehran, Iran



Skills

Python	80%
FineTuning TTS-ASR models	80%
Voice Cloning/Speaker Diarization	80%
Audio+Text Dataset Building	80%
FineTuning LLMs	60%
RAG/LangChain/Prompt Engineering	60%
API design/Flask/FastAPI	60%
Docker	60%
PostgreSQL/ChromaDB	60%
Git/Hugging Face	80%
linux/On-Prem/remote server	80%
nginx	60%
Selenium(Automation)	60%
Agile methodology	80%
Project management/Time Management	80%
Research/Documentation/Presentation skills	80%
interpersonal skills	80%
Computer Vision/Image Processing	60%



Work Experience

AI Engineer

Innovation Center of Cognitive Sciences and Technologies of IUST
(cognitive.iust.ac.ir)

Tehran, Iran

October 2023 - Present

Tasks and Achievements

- AI Project implementation in the fields of Speech and Text and Multimodal AI, including: keyword extraction from "audio and video" sources, Voice Cloning, Text-to-Speech (TTS) models, Speech-to-Text (STT) models, Building Audio+Text datasets, FineTuning TTS modes, Speaker Diarization, Ai video dubbing, Audio/video content recognition, music separation from audio sources, denoise and dereverberation, sentiment analysis from text and audio sources, working with LLMs, Prompt Engineering, Building dataset and Fine-tuning LMs, and working with remote servers(linux/shell), flask and fastapi, dockerize, nginx for deploying AI products.

AI Engineer

Shams company

Tehran, Iran

April 2023 - April 2024

Tasks and Achievements

- I am responsible for producing dubbing voice(Building dataset and training AI TTS models and voice cloning models) in different languages(English, Arabic and Persian). through this project, we went through all the steps from collecting audio data, cleaning it(preprocessing), building a dataset, and then training on it.

Product Owner & Analyst

Sadad Fars (Bale Messenger: www.bale.ai)

Tehran, Iran

August 2020 - January 2021

Tasks and Achievements

- I was in charge of managing the growth of the new product in Bale with Agile methodology : "Premium Content" Service.
- I was also in charge of identifying product problems through "testing" , "analysis" and "Marketing Research" and provided solutions to the technical team for implementation. For example, I was responsible for "Price Optimization" through testing and visualizing data and insights.

Osint Analyst

JameCo (jameco.ir)

Tehran, Iran

July 2018 - September 2019

Tasks and Achievements

- During this time, in addition to "Analysing Open Source Data" with Osint , I was responsible for setting the "Roadmap" for each Project.



Social Network



fatemeh-vahidunesi



F-V-Younesi



Projects

Intelligent video dubbing system: Dubbing video with "original voice" and "automatic subtitle creation"

For: Innovation Center of Cognitive Sciences and Technologies of IUST
August 2025

Building a multilingual video dubbing and subtitle creation service (including designing three separate algorithms for dubbing(Hard Sync, Soft Sync, Voice-Over), Voice Cloning modeling of individual voices, few-shot modeling of text-to-speech or TTS model, using speech recognition models, phrase-to-phrase translation model, noise removal model, Subtitle Creation , and music removal model from audio), API creation and Dockerization.

Building Interactive LLM powered NPC in a Metaverse Game(Integration of STT+Emotion Recognition from text and audio+ LLM response+ TTS model)

For: Innovation Center of Cognitive Sciences and Technologies of IUST
2024

This is a big Project as we build a multimodal AI Product(Interactive LLM powered NPC in a Metaverse space). this project needs Integration of 4 essential models:

- 1- ASR model: We Investigated different models for our purpose in different languages, and the worked on Streaming it for increasing speed.
- 2- Emotion Recognition from text and audio
- 3- LLM response: We used RAG and Prompt Engineering Methods for getting optimal and correct responds.
- 4- TTS model: We need certain person's voice, then we went through all the steps from collecting audio data, denoising and removing Reverb and background sounds, cleaning it(Preprocessing), building an audio dataset, and then training on it. then for optimizing speed, we worked on streaming TTS model.

Website content Reader(Voice Cloning + TTS)

For: Innovation Center of Cognitive Sciences and Technologies of IUST
2023

Keyword Extraction from Video/Audio sources

For: Innovation Center of Cognitive Sciences and Technologies of IUST
2024

Voice Cloning with so-vits-svc model + Building Audio Dataset

For: Shams Company
August 2023

Emotion Recognition from Tweet's text

For: Innovation Center of Cognitive Sciences and Technologies of IUST
2024

Emotion Recognition from arabic Tweet's text with Fine-tuning Bert model on specific Arabic dataset



Certificates

Audio Signal Processing

Institute: Coursera
2024

Link: <https://www.coursera.org/learn/audio-signal-processing>

Deep Learning

Institute: Coursera
2022

Link:

<https://www.coursera.org/account/accomplishments/verify/4YC6KPZXH8X>



Machine Learning

Institute: Sharif University of Tech

August 2021

PostgreSQL

Institute: Udacity

2021



Education

Master of MBA

Branch: Strategy

Institute/University: Sharif University of Technology

Iran

2020 - 2022

Bachelor of Industrial Engineering(Studying in two fields simultaneously)

Institute/University: Sharif University of Technology

Iran

2017 - 2018

Bachelor of Statistical Physics

Institute/University: Sharif University of Technology

Iran

2014 - 2018



Honors

Between the Top Ten Percent of CodeCup6 participants in "Data Analytics" Competition.

January 2022

Fourth place in "Osint(Open Source Intelligence)" field in "Fanavard National Competitions"

2019

Language

English					Arabic				
Reading					Reading				
Writing					Writing				
Speaking					Speaking				
Listening					Listening				